

Project Asteria - Sessions 6 & 7 Detailed Notes

Session 6: Motivation Engine

The Motivation Engine sits between the robot's long-term mission and its moment-to-moment decisions. The Primary Goal (for example, Explore Mars) never changes, but the Motivation Engine constantly evaluates what is most important right now. A low battery can temporarily override exploration; after charging, exploration becomes important again. Motivations are represented by priorities rather than fixed commands. Suggested motivations include: Stay Alive, Help Humans, Recharge, Explore, Protect Itself from Damage, Maintain Communication, Learn, Complete Mission, Return to Safe Location, and Follow Human Commands. Many ideas such as avoiding fire, avoiding water, slowing down on rough terrain, cliff avoidance, and cautious interaction with unknown objects are not motivations themselves. They are behavior modules that support higher-level motivations such as Stay Alive or Protect Itself. The architecture is: Sensors -> Sensor Fusion -> Environment Model -> Internal State + Memory -> Motivation Engine -> Priority Calculator -> Decision Engine -> Behavior Modules -> Motion Controller -> Motors. Priorities are dynamic. For example, Recharge may have priority 20 at 70% battery, 80 at 20%, and 100 at 5%. Internal variables such as Energy, Risk, Confidence, Curiosity, Damage and Mission Progress influence these priorities.

Session 7: Memory System

Memory allows Asteria to learn from the past instead of reacting as if every situation is new. Memory hierarchy: 1. Sensory Memory: current sensor readings lasting only fractions of a second. 2. Working Memory: information needed for the current decision. 3. Short-Term Memory: recent events and actions to avoid repeating mistakes. 4. Long-Term Memory: maps, objects, humans, hazards, successful routes, missions and learned knowledge. A dedicated Memory Manager stores, retrieves, updates, forgets and ranks confidence for memories. Object Memory Records should include identity, colour, shape, size, weight, texture, material, position, surroundings, environmental fingerprint, interaction history, safety assessment, confidence, timestamps, and relationships with other objects. One key design idea discussed was contextual memory. Instead of remembering only an object, Asteria remembers the environment around it. Surroundings act as an Environmental Fingerprint (walls, lighting, floor type, nearby objects, signal strength, temperature, etc.). This allows the robot to recognize places and infer whether it has likely encountered the same object or location before. Relationship memory stores links such as 'cup on table' or 'laptop near charger', making searching and planning faster. Event memory records the history of interactions with an object (seen, moved, picked up, scanned, returned). Predictive memory lets Asteria learn patterns over time, such as a door usually opening every morning at a particular hour. Finally, long-term memory is divided into Semantic Memory (facts about the world, e.g. charging station location) and Procedural Memory (skills and procedures, e.g. docking with the charging station). This separation allows the robot to acquire knowledge independently of learned skills and forms the foundation for future learning.